

Title	SCADA																																
Creator / Responsible Partner	CRF																																
Dataset Identifier	08_AnonymisedMESCRF I-BiDaaS WP2_D2.1 v0.1.7z																																
DOI	10.5281/zenodo.4265324																																
Dataset Description	The dataset contains sensors data of welding process in the sub-assembly lines of the vehicle. Different types of sensors are installed on the production line and acquire different data information (e.g. acceleration, velocity, pressure, temperature and so on). All of the sensors record their perception of the surroundings, uploading and transfer this information to a server that manages the data. For example, accelerometers are used for measuring vibration and shock on machines and basically anything that moves. Therefore, the monitoring of vibrations is important to check the status of a machine and the analysis of the trend of vibrations over time allows to predict the onset of deterioration and to intervene in time before the failure for the Predictive maintenance. This dataset can be used to analyse sensor data and obtain thresholds for anomalous measurements. There are 147 sensors, listed by categories as follows: <table><tr><td>Number of sensors</td><td>Physical quantity measured</td><td>Unit of measurement</td></tr><tr><td>87</td><td>Acceleration</td><td>mg</td></tr><tr><td>30</td><td>Velocity</td><td>mm/s</td></tr><tr><td>9</td><td>Temperature</td><td>°C</td></tr><tr><td>8</td><td>Pressure</td><td>bar</td></tr><tr><td>8</td><td>Flow</td><td>l/min</td></tr><tr><td>1</td><td>Displacement</td><td>mm</td></tr><tr><td>2</td><td>Energy vector (water)</td><td>l</td></tr><tr><td>1</td><td>Energy vector (air)</td><td>m³</td></tr><tr><td>1</td><td>Energy vector (air)</td><td>m³/h</td></tr></table> SCADA to obtain thresholds for anomalous measurements for all sensors			Number of sensors	Physical quantity measured	Unit of measurement	87	Acceleration	mg	30	Velocity	mm/s	9	Temperature	°C	8	Pressure	bar	8	Flow	l/min	1	Displacement	mm	2	Energy vector (water)	l	1	Energy vector (air)	m ³	1	Energy vector (air)	m ³ /h
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Work Package/Deliverable	WP2/D2.1																																
Source	Data have been retrieved from sensors mounted on several machines (e.g. linear stages, robots, elevators) along the production line of vehicles. We focused on welding lines in which robotic is used to assembly vehicles' components and flexibility is required for the continual changes of the types of components and vehicles.																																
Processing	Data have been processed following CRF rules																																
Repository	Zenodo																																
Language	English																																
Code list																																	

		Id	Value	Unit	Timestamp
	Example	667	49.75	mg	23/04/2018
	‘Id’ column reports the number that identifies the specific sensor. The other columns report on the value of the specific sensor, the unit of measurement and the timestamp.				
Type	Number				
Format	csv				
Expected Size	20.1 GB				
Keywords	sensor data				
Version	V0.1				
Date of Repository Submission	07/12/2020				
Necessary software	- any software enable to open formatted txt files (Notepad, Excel ...) is sufficient to view them - the software to analyse data depends on the type of analysis				
Rights	CRF				
Access Information	Open				